

Consider the linear optimization problem

$$\min_{x \in \mathbb{R}^2} x_1 + x_2$$

subject to

$$0 \leq x_2 \leq 1.$$

Write the problem in standard form, and consider the feasible basic solution where x_2 is in the basis. Calculate all the basic directions. Verify if they are feasible, and represent them on a figure.